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Health Outcomes and Economic Indicators: A Cross-Country Panel Study of Life Expectancy

GitHub

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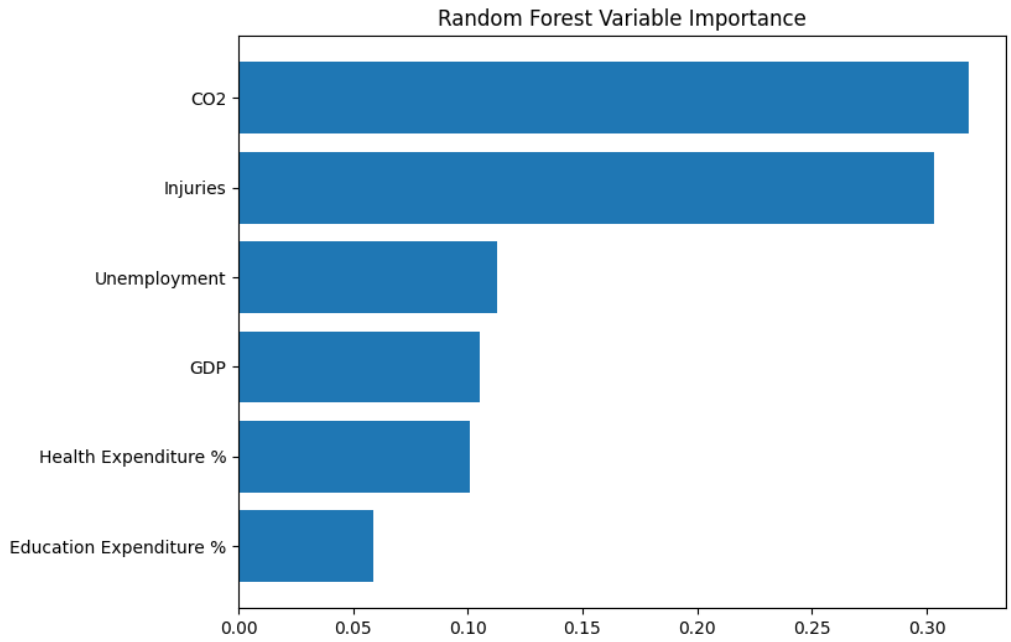
Executive summary

This project analyzes global life expectancy using country-level data from the World Bank for 166 countries between 2001 and 2019. We investigate the question: **"What economic, environmental, and health indicators are associated with life expectancy worldwide, and what are their relative effects?"** Using merged datasets, missing value imputation, and a Random Forest Regressor, we evaluate the predictive power of key variables and interpret their relationships through visualization.

Our major insight from this analysis is **CO_2 emissions, Injuries (measured as Disability-Adjusted Life Years, or DALYs), Unemployment (% of labor force), and GDP are the most important features in predicting life expectancy with varying effects.** GDP and CO_2 show positive relationships with diminishing returns, Injuries (DALYs) exhibits a strong negative relationship, and unemployment exhibits a nonlinear and more complex effect.

The variable importance plot is depicted below. Note the plot does not show directionality of feature importance, but rather the contribution each feature had to reducing prediction error in the model.



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Data sources

Both datasets used in this project were obtained from Kaggle, an online platform that provides public datasets, notebooks, and tools for data analysis. Although the datasets were downloaded from Kaggle, both were originally sourced from the World Bank.

- The first dataset focused on life expectancy and related socioeconomic and health indicators, including variables such as Life Expectancy, CO2 emissions, Health Expenditure, Education Expenditure, Unemployment, and disease burden measures. This dataset was published on Kaggle by the author MJShri and sourced from the World Bank. The dataset can be found here: https://www.kaggle.com/datasets/mjshri23/life-expectancy-and-socio-economic-world-bank.
- The second dataset contained GDP data by country and year and was used to incorporate economic indicators into the analysis. This dataset was published on Kaggle by the author zgrcemta and was also sourced from the World Bank. The dataset can be found here: https://www.kaggle.com/datasets/zgrcemta/world-gdpgdp-gdp-per-capita-and-annual-growths.

Data Preparation

Missing values were addressed using a two-step imputation process. First, linear interpolation was applied within each country over time. This method estimates missing values using surrounding observations, assuming gradual change across years. It is well-suited for panel data where economic and health indicators typically evolve smoothly. However, interpolated values are estimated and may reduce natural variability, potentially affecting model performance, and variable importance.

Countries with missing Life Expectancy for all years (2001–2019) were removed. As the response variable, Life Expectancy could not be reliably imputed, and retaining such countries would reduce model validity.

A second round of imputation was then performed using median values within each Income Group. This approach leverages similarities across comparable economies and improves data completeness. However, it may reduce variability by making countries within the same income group more similar, potentially weakening country-specific effects.

Most variables had low missingness (generally <10%). The main exception was Education Expenditure (~33% missing), which was retained after imputation. Results involving this variable should be interpreted with caution, as extensive imputation may affect its reliability and influence in the model.

Method of Analysis

In order to address the research question regarding which factors contribute best to life expectancy predictions for different countries, Random Forest Regressor was used. This method was chosen because relationships between variables (such as GDP, CO_2, and health indicators) were not linear, making traditional regression unsuitable.

The data was first cleaned and prepared by merging datasets, handling missing values through interpolation and income-group imputation, and removing countries with no life expectancy data.

The model was trained using country-level splits to avoid data leakage, and performance was evaluated using cross-validation to ensure reliable results. Variable importance was used to identify which factors matter most, and Partial Dependence Plots (PDPs) were used to understand how those factors affect life expectancy.
(See Appendix for more details on methods)

Findings

Random Forest Model Performance

- **$R^2 \approx 0.68$: This means the model explains 68% of the variation in life expectancy.**
- **RMSE ≈ 5.14 years: The means the models predictions are off about 5.14 years of life expectancy on average.**
- **Average Cross-validated $R^2 \approx 0.58$: across different splits of the data, the model explains on average 57.8% of variation.**

  This indicates that the model captures key patterns and explains a substantial proportion of the variation, though some variation remains unexplained. This is expected given the complexity of global life expectancy and data and the models performance is reasonable for this type of cross-country, real-world data.

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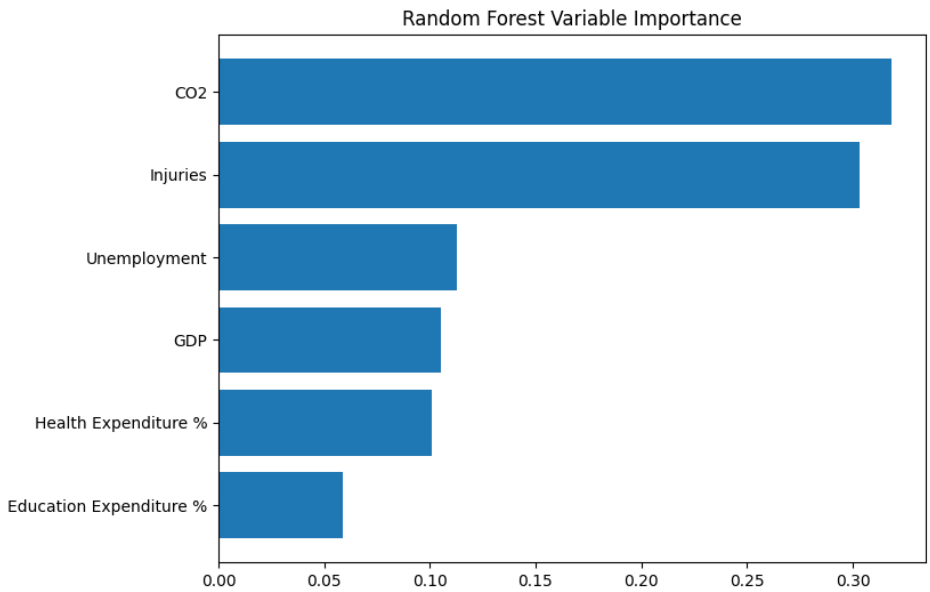
Variable importance

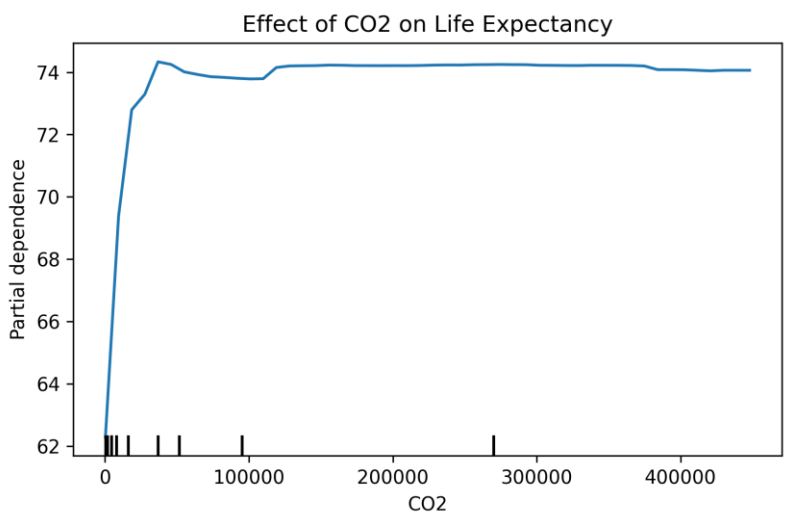
Variable importance results show that the strongest predictors are in order of importance: **CO_2 emissions, Injuries (DALYs), Unemployment, GDP.**



Variable importance identifies which factors matter most, but not how they affect life expectancy. To understand direction and relationships, PDPs were used *(see Appendix for interpretation details)*.

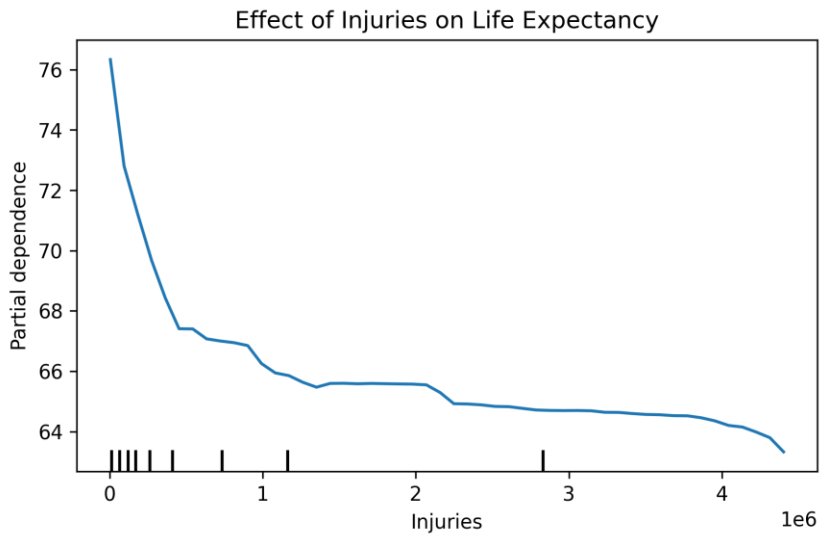
Feature relationships (PDP)

CO_2 v Life Expectancy



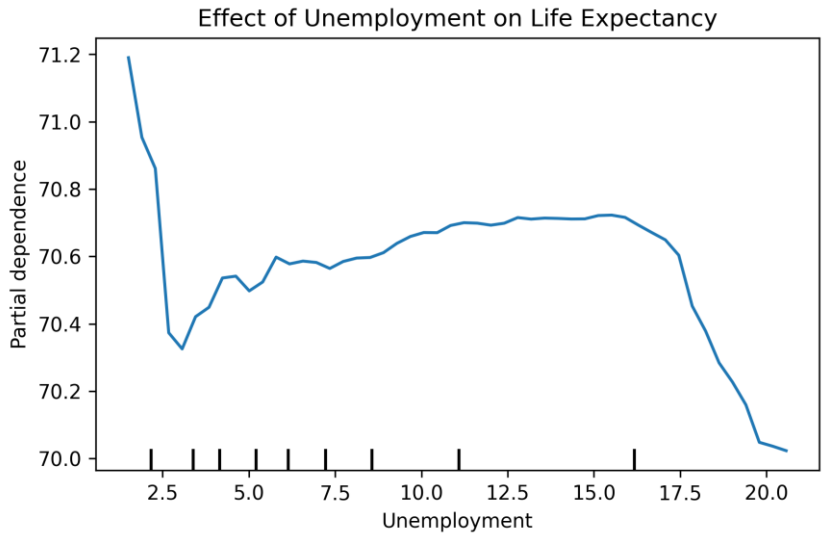
A sharp rise in predicted life expectancy at low levels of CO2, followed by a plateau, indicating a positive, nonlinear diminishing returns relationship. One explanation could be proxy effect: countries with higher CO_2 emissions tend to be more industrialized, wealthier, and have better healthcare systems.

Injuries (DALYs) v Life Expectancy



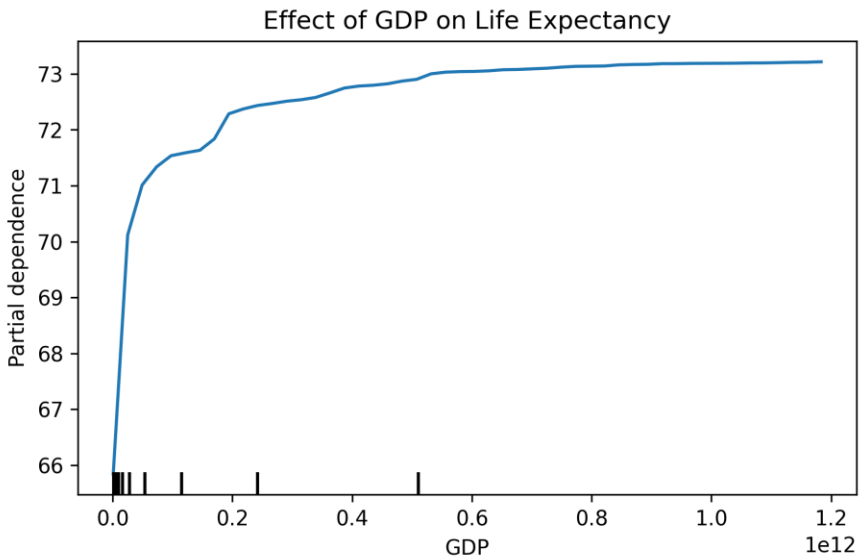
Strong nonlinear, negative relationship, where increases in injury-related DALYs are associated with a sharp decline in predicted life expectancy, followed by a more gradual decrease at higher levels.

Unemployment v Life Expectancy



Nonlinear and complex effect-very low unemployment depicts higher life expectancy and very high unemployment with lower life expectancy. A big dip at lower levels and then plateau at moderate levels depicting more stable conditions.

GDP v Life Expectancy



Strong positive relationship with diminishing returns: Substantial gains in life expectancy occur at lower GDP levels, followed by a plateau at higher levels. Early increases likely reflect improvements in infrastructure, healthcare, and living conditions, while additional growth yields smaller gains.

Appendix

Random Forest

Random Forest is a machine learning method used to make predictions by combining many individual decision trees. A **decision tree** works like a flowchart. It asks a series of simple questions, such as:

- Is GDP high or low?
- Is health expenditure above average?

Based on the answers, the tree arrives at a prediction for life expectancy. Rather than relying on just one tree, Random Forest builds **hundreds of trees**, each using slightly different samples of the data. Cross-validation is a method used to test how well a model performs on new, unseen data. Instead of evaluating the model on a

single sample, the data is repeatedly divided into different training and testing groups, and the model is evaluated multiple times.

This is important because it provides a more reliable measure of performance and helps ensure the model is not just fitting patterns specific to one dataset (overfitting). Cross-validation was done at the country level, meaning the model was tested on entirely unseen countries, giving a more realistic assessment of how well it generalizes. (From earlier analysis, splitting the data on country-year had the same counties in training and test set leading to data leakage. The model was performing too well because it was using specific countries' previous data to predict a different year.) Every tree makes its own prediction, and the model **averages all predictions together**.

This approach:

- Improves accuracy
- Reduces the influence of unusual data points
- Captures complex relationships between variables

Variable Importance

Random Forest provides **variable importance scores**, which measure how much each predictor contributes to reducing prediction error across all trees.

- Higher importance = the model relies more on that variable
- Lower importance = the variable contributes less to predictions

Importantly, **variable importance does NOT show directionality**.
It tells us *which variables matter most*, but not whether they increase or decrease life expectancy.

Partial Dependence Plots (PDPs)

To understand how each variable affects life expectancy, we use **Partial Dependence Plots (PDPs)**.

A PDP shows:

- The value of a predictor (x-axis)

- The model's predicted life expectancy (y-axis)

These plots illustrate the **direction and shape of the relationship**, helping answer:

How does a change in this variable effect life expectancy?

Unlike variable importance, PDPs **do show directionality**, making them essential for understanding the effects of each predictor.